

Empirical Project

Machine Learning in Econometrics

due on July 31, 2026*

1 Goal

Many cities around the world have introduced policies aimed at reducing greenhouse gas emissions from road transport. Two increasingly common policies are

- **congestion pricing**, which charges vehicles for entering certain parts of a city, and
- **low-emission zones**, which restrict access for highly polluting vehicles.

Both policies have become central elements of urban climate policy, yet there remains considerable uncertainty regarding their effectiveness. A group of politicians from cities who have implemented at least one of the policies are interested in the following questions:

1. Were the policies effective in reducing emissions?
2. Which type of cities benefit the most/least from the policies?
3. Does implementation of both policies simultaneously produce greater emission reductions than what would have been expected from the sum of the individual effects of implementing either policy alone?

*to be submitted on moodle by 23:59 German time

The politicians hire you to produce a report that addresses these questions based on a panel dataset containing annual observations for 183 anonymized cities from 2008 to 2024. To answer the questions, the politicians ask you to estimate suitable causal effects of the two policies and to use your machine learning knowledge to flexibly control for possible confounding variables.

2 Data

The dataset for this project is called `urban_emissions_panel` and is available in csv and RData formats on moodle. A separate file, `variable_descriptions.csv`, provides detailed descriptions of all variables in the dataset.

The dataset contains annual observations for anonymized cities. The main variables are:

variable name	description
<code>city_id</code>	Unique identifier for a city
<code>year</code>	Calendar year
<code>transport_co2</code>	Annual CO ₂ emissions from road transport (kt)
<code>total_co2</code>	Annual total city-wide CO ₂ emissions (kt)
<code>cp_active</code>	Indicator: congestion pricing active in this year (1 =yes)
<code>lez_active</code>	Indicator: low-emission zone active in this year (1 =yes)
<code>cp_impl_year</code>	Year congestion pricing was first implemented (0 if never)
<code>lez_impl_year</code>	Year low-emission zone was first implemented (0 if never)
<code>cp_announce_year</code>	First year congestion pricing became a major topic in local public discussions (0 if never)
<code>lez_announce_year</code>	First year low-emission zone became a major topic in local public discussions (0 if never)

In addition to these variables, the dataset contains a rich set of annual city characteristics. For further information on all variables, consult `variable_descriptions.csv`.

3 Instructions

3.1 General instructions

Write a report that describes your analysis and findings. It should read like an applied empirical research paper that you could send to the group of politicians described above, not like a programming exercise or a list of software outputs.

Throughout the report, justify your methodological choices and assumptions. Interpret your findings critically and discuss the strengths and limitations of your empirical strategy.

It is very important that your report is convincing, precise, and concise. Spend a substantial amount of time polishing the report so that it is easy to read, carefully proofread, and so that tables and figures are clearly labelled and easy to interpret.

3.2 Report structure

Your report must contain the following sections.

1. **Introduction.** Describe the policy question, why it is economically relevant, and the objectives of your empirical analysis.
2. **Empirical strategy and identification.** Define your target parameters and provide assumptions under which they are identified.
3. **Proposed estimator.** Propose an estimator and provide a step-by-step explanation of how to implement it, including tuning parameters and any other implementation choices.
4. **Theoretical properties.** Explain why your estimator is expected to perform well and state the main assumptions required for consistent estimation and valid inference.
5. **Data preparation.** If you process the data before applying your proposed estimator, e.g. by dropping observations or transforming variables, describe such operations.
6. **Results.** Present your main empirical findings. Include appropriate tables, figures, estimated effects, and confidence intervals. Select the most informative results rather

than reproducing large amounts of software output. Supporting evidence can be included in the appendix.

7. **Sensitivity analysis.** Investigate the robustness of your conclusions to the various modeling choices made in your analysis.
8. **Interpretation and discussion.** Discuss the effectiveness of the policies and the strengths and limitations of your empirical analysis.

3.3 Theory requirements

Any assumptions, concepts, estimators, and results not covered in the lecture notes or problem sets of the course “Machine Learning in Econometrics” must be introduced and explained carefully. You should not assume that the reader is familiar with material from outside the course. Proofs and derivations are not required with one exception: if you claim Neyman orthogonality of your target parameter, you need to formally prove this property.

3.4 Permissible Resources

You must work on this project on your own; no group work is allowed. You must produce the report as well as the code on your own. You may use any “non-human” resources available to you, i.e. books, information on the internet, and the lecture materials. However, you are not allowed to use interactions with other humans, in particular no discussions with your classmates about the project, no discussions with others through the internet, and no soliciting of help in online forums etc. are permitted.

You may use AI tools like ChatGPT, but be careful to make sure that you do not lose authorship of the report: if large parts of your report or code are generated by AI, then you may not be considered to be the author anymore. Make sure that you use such tools only to help you with specific, small tasks (e.g., spell-checking, improving the writing, helping with formatting a plot or a table).

3.5 Format of the Report

- Length: max. 10 pages of main text, including figures, tables, and references

- Appendices of up to 10 additional pages are allowed. They may contain additional derivations, supplementary figures, robustness analyses, implementation details, or additional empirical results. Material essential for understanding the main results should remain in the main text
- The report has to be a PDF document
- Filename: `[lastname]_[firstname]_[studentnumber].pdf`
- Take time to polish the text, so it is easy to read, precise and concise.
- The report must include the following sentence at the end:

“I confirm that this report is based on my own work. In preparing this report, I have not received any help from another human nor have I discussed any aspects of the empirical project with others.”

Then sign the report just below this statement.

3.6 Code

You must submit all code used in your statistical analysis in a separate file or multiple files. Make sure your code is readable and well-documented.

3.7 Expected Workload

As a rough guideline, the expected amount of time spent on the empirical project is: 2 hours for the theoretical analysis + 15 hours of empirical analysis + 3 hours of carefully writing up the report. This leads to a total of about 20 hours.